WEEKEND ENGINEERING ASSIGNMENT

15-Day Intern Performance Evaluation | Weekend 1 | HR distribution copy

Purpose. This assignment forms part of a 15-working-day assessment of independent project understanding, system design, testing, technical decisions, implementation and ownership. Each intern completes all three projects independently alongside their weekday client-project responsibilities.

1. Projects for this weekend

Project	Minimum functional scope
Blog Platform	Authentication, author/admin roles, create/edit posts, draft/publish, categories, comments and search.
E-commerce Store	Product catalog, cart, stock control, mock-payment checkout, orders and order status.
Two-sided Marketplace	Buyer/seller accounts, listings and search, order or offer flow, and reviews.

Scope: Build a coherent working MVP for each project. Prioritize a complete end-to-end user journey. Identify omissions and reasons; never represent incomplete functionality as implemented.

2. Mandatory work for EVERY project

• Requirements and flow: Define users, business goals, scope, assumptions, roles, acceptance criteria and complete user journey.

• Architecture: Diagram client, server/API, data storage, authentication, integrations and background services; show data movement and deployment locations. Justify alternatives.

• Data design: Provide ERD, schema, primary/foreign keys, constraints, relationships, indexes and data-integrity considerations.

• API design: Provide sequence/API flows for success and failure; routes, methods, auth, request/response examples and error cases.

• Implementation: Provide runnable MVP, setup steps, environment variable names (no secrets), and demo.

• Self-testing: Write and execute positive, negative, boundary, permissions, duplicate/retry and regression tests with expected and actual results.

• Deployment and commercial analysis: Compare at least two realistic hosting options (e.g., VPS/Hostinger and AWS). Estimate compute, database, storage, backups, bandwidth/egress, monitoring, domain, third-party and worker costs; state region, currency, assumptions, vendor source URLs and access dates. Explain Docker, CI/CD, queues, caches, security and observability, distinguishing implemented versus proposed.

• Scaling: Explain design changes, bottlenecks, database/storage/queue changes and cost at 100, 1,000, 10,000 and 1,000,000 users; state traffic assumptions.

• Open-source and AI policy: Building from scratch or adapting open source is allowed; disclose repo URL, license, reused parts, modifications, and original work. AI use is not encouraged. If used, disclose tool and assisted areas and verify independently. All outputs must be defensible in interview.

• Submission and packaging: Submit one accessible Drive parent folder Weekend1_YourName with 01_Blog, 02_Ecommerce and 03_Marketplace. Each project includes Section 3 deliverables or clearly named links, editable Lucidchart links, exported PDFs, repo and demo links, and reviewer access. No credentials/secrets.

• Review and interview: Assigned Friday, completed over weekend, submitted before Monday interview to HR and assigned Junior 1/2 reviewer. Reviewer may question all design, cost and testing decisions and request walkthrough, debugging or small changes.

3. Exact deliverable formats — for EACH project

File / output	Format and required contents
01 - README.md	Purpose, tech stack, scope, assumptions, architecture summary, exact setup/run/test instructions, known gaps.
02 - architecture.pdf	Lucidchart-exported system architecture; label components, data movement, integrations and deployment boundaries. Include editable Lucidchart link.
03 - database-design.pdf	Lucidchart ERD with cardinality, PK/FK, indexes and field types; include schema.sql or migrations and editable link.
04 - api-workflows.pdf	Lucidchart sequence/API diagrams: one complete success path and failure path; include api-spec.md or openapi.yaml.
05 - test-cases.xlsx	Columns: Test ID, Feature, Preconditions, Steps, Input, Expected Result, Actual Result, Status, Defect ID, Evidence Link; include execution proof.
06 - engineering-decisions.md	Options considered, final choices, rationale, trade-offs, security, technical risks, implemented vs planned services and remediation of known gaps.
07 - source + demo	Runnable GitHub repository with README plus an accessible 3–5 minute demo recording/link for each project.
08 - cost-and-scaling.xlsx	One comparison table per hosting option: service, specification, region, usage assumption, unit price, monthly estimate, source URL and access date. Totals and scenario estimates for 100 / 1k / 10k / 1M users. Mark assumptions vs verified figures.
09 - sources-and-disclosure.md	List open-source repository links/licenses and exactly what was reused/modified. State AI tools used, the work assisted, verification undertaken, and citations for external pricing or technical claims. Write “None” where applicable.
10 - submission-index.md	Inside the parent folder, list all three project subfolders, repository/demo links, editable Lucidchart URLs, PDF exports, test evidence and any incomplete deliverables. Confirm reviewer access.

Interview readiness: No extra file is required for the Monday interview; review your diagrams, code, tests, pricing sources, assumptions and trade-offs. The same assigned reviewer will question these outputs.

4. Evaluation and performance intent

Dimension	Weight
Problem understanding	20%

Dimension	Weight
Architecture and design	25%
Implementation quality	25%
Testing and edge cases	20%
Explanation and independent ownership	10%

Each project is assessed separately from 1–5 in each dimension. Weighted score = sum of (rating / 5 × weight). The weekend result contributes to the broader 15-day performance evaluation. Genuine understanding, credible evidence and independent judgment matter as much as functioning code. Disclose unfinished items, blockers and next steps.